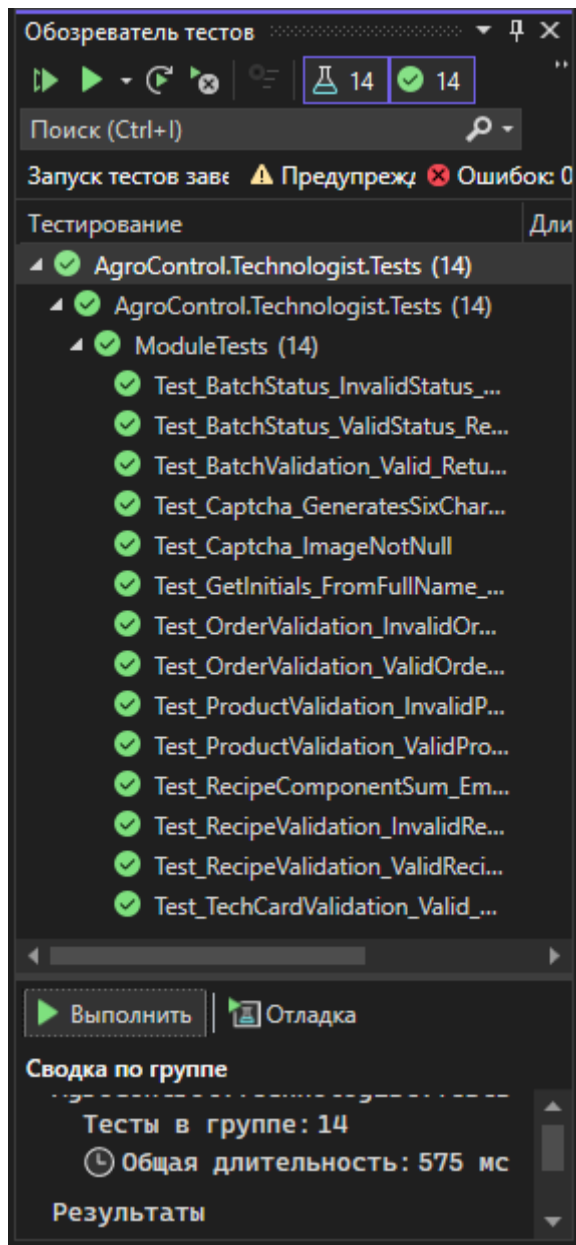




```
using Microsoft.VisualStudio.TestTools.UnitTesting;
```

```
using AgroControl.API.Models;
```

```
using AgroControl.Techonologist;
```

```
using System.Collections.Generic;
```

```
namespace AgroControl.Techonologist.Tests
```

```
{
```

```
    [TestClass]
```

```
    public class ModuleTests
```

```
    {
```

[TestMethod]

```
public void Test_Captcha_GeneratesSixCharacters()
{
    string captcha = TestHelpers.GenerateCaptchaText();
    Assert.AreEqual(6, captcha.Length, "Капча должна быть длиной 6 символов");
}
```

[TestMethod]

```
public void Test_Captcha_ImageNotNull()
{
    string text = "ABC123";
    bool isGenerated = TestHelpers.IsCaptchaImageGenerated(text);
    Assert.IsTrue(isGenerated, "Изображение капчи должно создаваться");
}
```

[TestMethod]

```
public void Test_ProductValidation_ValidProduct_ReturnsTrue()
{
    var product = new Product { Код = "P001", Наименование = "Гербицид" };
    bool isValid = TestHelpers.IsProductValid(product);
    Assert.IsTrue(isValid);
}
```

[TestMethod]

```
public void Test_ProductValidation_InvalidProduct_ReturnsFalse()
{
    var product = new Product { Код = "", Наименование = null };
    bool isValid = TestHelpers.IsProductValid(product);
    Assert.IsFalse(isValid);
}
```

```
}
```

```
[TestMethod]
```

```
public void Test_RecipeValidation_ValidRecipe_ReturnsTrue()
```

```
{
```

```
    var recipe = new Recipe { ПродуктID = 1, Версия = 2 };
```

```
    bool isValid = TestHelpers.IsRecipeValid(recipe);
```

```
    Assert.IsTrue(isValid);
```

```
}
```

```
[TestMethod]
```

```
public void Test_RecipeValidation_InvalidRecipe_ReturnsFalse()
```

```
{
```

```
    var recipe = new Recipe { ПродуктID = 0, Версия = 0 };
```

```
    bool isValid = TestHelpers.IsRecipeValid(recipe);
```

```
    Assert.IsFalse(isValid);
```

```
}
```

```
[TestMethod]
```

```
public void Test_OrderValidation_ValidOrder_ReturnsTrue()
```

```
{
```

```
    var order = new ProductionOrder { НомерЗаказа = "ORD-001", ПланКоличество_кг =  
100 };
```

```
    bool isValid = TestHelpers.IsOrderValid(order);
```

```
    Assert.IsTrue(isValid);
```

```
}
```

```
[TestMethod]
```

```
public void Test_OrderValidation_InvalidOrder_ReturnsFalse()
```

```
{  
    var order = new ProductionOrder { НомерЗаказа = "", ПланКоличество_кг = 0 };  
    bool isValid = TestHelpers.IsOrderValid(order);  
    Assert.IsFalse(isValid);  
}
```

```
[TestMethod]  
public void Test_TechCardValidation_Valid_ReturnsTrue()  
{  
    var card = new TechCard { ПродуктID = 1, Версия = 1 };  
    bool isValid = TestHelpers.IsTechCardValid(card);  
    Assert.IsTrue(isValid);  
}
```

```
[TestMethod]  
public void Test_BatchValidation_Valid_ReturnsTrue()  
{  
    var batch = new ProductionBatch { НомерПартии = "B-001", ЗаказID = 5 };  
    bool isValid = TestHelpers.IsBatchValid(batch);  
    Assert.IsTrue(isValid);  
}
```

```
[TestMethod]  
public void Test_GetInitials_FromFullName_ReturnsCorrect()  
{  
    string fullName = "Иванов Иван Иванович";  
    string initials = TestHelpers.GetInitials(fullName);  
    Assert.AreEqual("ИИ", initials);  
}
```

[TestMethod]

public void Test_BatchStatus_ValidStatus_ReturnsTrue()

```
{  
    bool isValid = TestHelpers.IsValidBatchStatus("выполняется");  
    Assert.IsTrue(isValid);  
}
```

[TestMethod]

public void Test_BatchStatus_InvalidStatus_ReturnsFalse()

```
{  
    bool isValid = TestHelpers.IsValidBatchStatus("неизвестный");  
    Assert.IsFalse(isValid);  
}
```

// Добавьте 10-й тест, если нужно – например, проверка суммы процентов рецепта

[TestMethod]

public void Test_RecipeComponentSum_EmptyComposition_ReturnsZero()

```
{  
    var recipe = new Recipe { Состав = new List<RecipeComponent>() };  
    decimal sum = TestHelpers.SumComponentPercentages(recipe);  
    Assert.AreEqual(0, sum);  
}  
}  
}
```